

MINILISTIC MNIST UTILIZING MODERN OPTIMIZATIONS

CS 211 #18849

Brendan Tea, brendan.tea@bellevuecollege.edu,
 Advisor: Professor Dr. Taesik Kim

Abstract: Value nodes with elementary operations can be built to create complex functions that achieve practical appliances. Although more memory intensive, this serves as a bare minimum for artificial intelligence architectures—that is without the computational optimizations. These value nodes hold a history of past operations and children nodes, creating a directed acyclic graph that allows us to calculate local gradients. Through these value nodes, we build up neurons, layers, and all the way up to a multilayer perceptron. With the MNIST dataset, we're able demonstrate the functionality of the value node multilayer perceptron using a training loop with gradient descent resulting in empirical results of 92~% accuracy. Utilizing optimizations such as L2 normalization, adaptive moment estimation (Adam), cross entropy loss, and Gaussian error linear unit (GELU) resulted to lower loss and a final accuracy of 97~%.

1 Introduction

Instead of using matrices and linear algebra to replicate a multilayer perceptron, we use value nodes. We define each fundamental operator—add, multiply, exponentialization, and trigonometric functions—alongside their partial derivatives with the chain rule. These individual value nodes will additionally hold previous children, creating a directed acyclic graph. All of this together recreates an autograd engine vital for back propagation. In whole, we can create a multilayer perceptron, train it on the MNIST dataset, and obtain promising results.

2 Building Up

With these operators, we can define any function atomically. To build up to a multilayer perceptron. A neural network is composed of layers which are composed of neurons. This structure (with a non-linear function) allows the approximation of any shape it or desired behavior.

DEFINITION 2.1 (Neuron). A neuron or perceptron is denoted by the following:

$$N = \sum_{i=0}^n w_i x_i + b, \quad n \text{ being the input size.}$$

$$N = \begin{bmatrix} x_0 & x_1 & \cdots & x_i \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_i \end{bmatrix} + [b],$$

i being the number of inputs.

DEFINITION 2.2 (Layer). A layer of neurons or a single layer perceptron is denoted by the following:

$$L = \begin{bmatrix} N_0 \\ N_1 \\ \vdots \\ N_i \end{bmatrix}, \quad i \text{ being the amount of neurons.} \quad (2.1)$$

DEFINITION 2.3 (MLP). A multi layer perceptron has multiple layers and it is denoted by the following:

$$\text{MLP} = [L_0 \quad L_1 \quad \cdots \quad L_i]$$

$$= \left[\begin{bmatrix} N_0 \\ N_1 \\ \vdots \\ N_a \end{bmatrix} \quad \begin{bmatrix} N_0 \\ N_1 \\ \vdots \\ N_b \end{bmatrix} \quad \cdots \quad \begin{bmatrix} N_0 \\ N_1 \\ \vdots \\ N_c \end{bmatrix} \right],$$

a, b, c being the size of each layer.

Now that we have some infrastructure to work with, we have to show how forward pass would work. Imagine we have a 2 layer (1 input/middle, 1 output). For the input layer we shall take in two inputs and output 4. The output layer must take 4 and we will let it output 1.

This is how the math would work out.

$$X_0 = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix}, L_0 = \begin{bmatrix} N_0 \\ N_1 \\ N_2 \\ N_3 \end{bmatrix}, N_i = \begin{bmatrix} x_0 & x_1 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} + [b]$$

$$L_1 = [N_0], N_0 = \begin{bmatrix} N_0 \\ N_1 \\ N_2 \\ N_3 \end{bmatrix}^\top \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ w_3 \end{bmatrix} \\ = [N_0 \quad N_1 \quad N_2 \quad N_3] \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ w_3 \end{bmatrix} + [b]$$

basically linear regression but on steroids. We will forward pass inputs and receive an output. With the output, we will calculate the loss and run back propagation for the individual gradients and apply gradient descent on the parameters to reduce the loss value. In turn, this will result in a model that is better suited toward the result. Repeating this process will eventually lead to any function reaching a minimum.

2.1 Back Propagation

To actually train and get the wanted behavior from a function such as a MLP, we must obtain the direction of largest or smallest growth otherwise known as the gradient of the parameters. With the gradient, we can use that to move in a direction of lower loss. With a little bit of multi-variable calculus and the chain rule, we can obtain all of the local derivatives and in turn, their gradients. Here is the math behind it.

The most rudimentary case is to imagine taking derivatives at the root.

$$\frac{\partial L}{\partial c} = \frac{\partial L}{\partial c} \overset{1}{\frac{\partial L}{\partial c}} = \frac{\partial L}{\partial c} \quad (2.2)$$

However, imagine if we have a function of two variables with one being a composite. In this example, it should become clear that we will always calculate the outer derivative and multiply it by the inner derivative. At the start, we will simply take it's derivative and multiply it by one.

DEFINITION 2.4 (Chain Rule). With that, back propagation is defined by the chain rule.

$$L(c, d) = c + d, \quad c(a) = a + b; \quad (2.3)$$

$$\Rightarrow \boxed{\frac{\partial L}{\partial a} = \frac{\partial L}{\partial c} \frac{\partial c}{\partial a}} \quad (2.4)$$

Accumulating gradients:

$$L(c, d) = c(a) + d(a), \quad c(a) = a + b, \quad d(a) = a + c \quad (2.5)$$

$$\Rightarrow \boxed{\frac{\partial L}{\partial a} = \frac{\partial L}{\partial c} \frac{\partial c}{\partial a} + \frac{\partial L}{\partial d} \frac{\partial d}{\partial a}} \quad (2.6)$$

DEFINITION 2.5 (Gradient). We can take this farther and define the gradient of the loss function.

$$\boxed{\nabla L = \left\langle \frac{\partial L}{\partial a_0}, \dots, \frac{\partial L}{\partial a_i} \right\rangle} \quad (2.7)$$

This is a very powerful tool and the next thing to tackle is how we order our function so that we can create an algorithm to start from the root and propagate back.

2.2 Topological Sort

With back propagation, we will need to be able to know what order to propagate. This is basically a job scheduling problem so topological sort will be the perfect algorithm for this. Topological sort works by applying a post-order depth first search on a directed acyclic graph and reversing the list. This can easily be done using a recursive formula to completely search each branch. Reversing should be extremely simple with a library.

Algorithm 2.1 Topological Sort

Require: $n \geq 0$

Ensure: $y = x^n$

```

 $y \leftarrow 1$ 
 $X \leftarrow x$ 
 $N \leftarrow n$ 
while  $N \neq 0$  do
  if  $N$  is even then
     $X \leftarrow X \times X$ 
     $N \leftarrow \frac{N}{2}$  {This is a comment}
  else if  $N$  is odd then
     $y \leftarrow y \times X$ 
     $N \leftarrow N - 1$ 
  end if
end while

```

3 MNIST

3.1 The introduction

As mentioned above, the introduction and conclusion together must give a complete and comprehensible picture of your thesis. This is for the time-poor readers whose interest has been aroused by the abstract but do not have time to read the full thesis,

or who in any case need to decide whether they want to devote time to reading it.

Your introduction should therefore set out the context of the work (why it is relevant, who has worked on the subject before), the specific question (with an explanation of how it contributes to our understanding of the subject) and a summary of the techniques and analysis methods used. Obviously, the introduction is not the place for a highly detailed discussion of the methods and techniques. In the case of long texts in particular, the discussion of the context and related work may be spread over several chapters. This does not alter the fact that it should be possible to read the introduction and the conclusion on a ‘stand-alone’ basis.

The introduction should convince the reader that your work is useful and exciting/interesting, that you know what you are talking about and understand why you researched the subject in the way you describe.

Introductions often end with a summary of what follows in the rest of the thesis. This is only useful if it makes the thesis easier to read. For example, it is useful to point out that Chapter 3 is only relevant for people who are interested in the mathematics of your model, and others therefore do not need to read it. It is not useful to point out that Chapter 2 describes the model, Chapter 3 contains the results and Chapter 4 the conclusion. The reader can see this from the table of contents. In the case of bachelor theses it is not necessary at all to point it out, as such theses are relatively short.

3.2 The system / methods

The description of the system and/or method must focus on reproducibility. In the case of a computer program, sufficient detail must be provided to enable the programming to be repeated. In the case of an experimental setup, this means that sufficient details should be given so that the precise experiment can be repeated.

For experimental papers, there is a much used, and therefore more or less standard structure. It is highly recommended to adhere to such a standard structure. It helps the reader because they know where to look for particular information, and it helps you as a writer because it is easier not to forget things.

In both cases, only relevant details should be given. Just as, in the case of an fMRI experiment, the colour of the pyjamas worn by the test subjects is unimportant but the type and settings of the MRI

machine *are* important, in the case of programs it is important to describe precisely *what* happens, but not precisely *how* this was programmed. The description therefore needs to be above the source-code level. Often, the best way is to describe in general how the system works and then discuss its individual components. Ultimately, everything must be substantiated with formulae and tables of pseudo code.

The reasons behind certain choices in the system should also be given. In many cases, certain aspects of the system can be executed in many different ways. It is therefore necessary to explain why a certain method of execution was chosen and why the alternatives were rejected. It is helpful to place these choices in the wider context of the research, as outlined in the introduction and research question. Describing the system therefore involves more than simply describing what you built!

Importantly, remember to include the values of all the parameters used in your system! The best way to do this is to present them in a table in a central position.

The description in the system/method section goes beyond the description of the computer model or experiment that was used. The techniques used to analyse the behaviour of the system (or the model, or the experimental subjects) should also be described. In many cases, a computer model or an experiment generates sizeable volumes of data that, in their raw form, are too large and complex to draw a conclusion from. In such cases, measures need to be defined for calculations using the data. These may be averages, or aspects such as the diversity of the system, learning success, rate of convergence, etc. The calculations must be clearly described, and reasons given for the choice of measures. For experiments, this means that the method section also describes how the raw data were collected, averaged, and processed to make it ready for statistical testing. The statistical tests must also be described, together with the reasons to use those.

3.3 The results

When you present your results, you are presenting the essence of the message you wish to convey. This is the opportunity to show what you have discovered and how your findings shed light on the research question. Two things are important in order to do this successfully: selection and presentation.

The importance of selecting results is something that young scientists/academics often overlook. Surely every result is important? Yes and no. It is

natural to want to present every single result that you have obtained from your model with blood, sweat and tears. The risk, however, is that you will overload the reader with results that are not relevant to your research question. It is therefore important to present only the results that shed light on your research question.

This does not alter the fact that not all your interesting findings relate directly to the research question. There are plenty of studies that have produced different and much more interesting results that do not relate to the original purpose of the study. But that is something for your next publication. The research for your thesis had a specific aim and, given the limited amount of time available, it is sensible to keep the focus on the research question you formulated at the outset.

The second problem is how to present the results. It goes without saying that they should be presented as clearly as possible. Unfortunately, it is not possible to say briefly which format is suitable in each case. However, there are certain guidelines that may be followed.

Graphs are useful for presenting large volumes of data. If there is a relationship between the data (e.g. data points in a time series), a line graph should be used. If no such relationship exists, a bar chart is preferable. If the results consist of components that add up to a whole, it is a good idea to present them in a pie chart, and sometimes it is useful to think up a graph of your own. A number of conventions for producing decent graphs are discussed in Section 4.7. As with writing, when you are learning to produce graphs it is useful to look critically at a wide range of graphs produced by other people.

Graphs are not the answer to everything. Sometimes it is better to present results in a table. This is certainly the case if precise numeric values are important, or in the case of very small numbers of data points. Sometimes it is sufficient to discuss the results in the text, certainly if the qualitative form of the results is more important than the precise quantitative values.

In all cases, you must show whether or not your results are statistically significant. In graphs this is done using box plots, reliability intervals or error bars. In tables it is useful to give reliability intervals and/or standard deviations, etc. When discussing your results you must also clearly explain which statistical tests you used and why they are valid tests to use.

3.4 The conclusion and discussion

In the conclusion and discussion you describe the consequences of your findings, obviously in the context of your research question. As mentioned above, the introduction, together with the conclusion and discussion, must form a mini-article that can be read on a stand-alone basis. Therefore you should somehow include all your key findings in this section. Clearly, there is no need to repeat graphs and tables, but it is essential to summarize your findings in words.

The repetition will occur when you explain what the results mean and how they should be interpreted. You should take the reader by the hand, as it were, by explaining how each result is relevant to the research question and to other work on the subject. This discussion may serve as an argumentation for a short and (hopefully) powerful message: the ultimate conclusion. Obviously, there may be more than one conclusion that can be drawn from your results.

In the discussion, the conclusion can be placed in the broader context of the academic discussion to which the presented work contributes. This is also the place to make suggestions for future research.

Possibilities for future research are sometimes (but not usually in a Bachelor's thesis) presented in a separate section. In this section, keen young academics at the start of their career have the tendency to suggest how the system can be made even more complex, especially if their results were disappointing. But remember that the reverse may be true: perhaps the system was too complex, and simplification would provide greater insight. Suggestions like these should always be made with the original research question in mind, not simply because it is enjoyable to play with complex systems. When you've done an experiment, don't simply suggest to use more subjects in a next experiment to increase the power of the experiment. Really large effects in the real world should have shown in your experiment even with a relatively small sample. Don't confuse significance with relevance!

3.5 Acknowledgements and related matters

Acknowledgements etc. are usually only appropriate in texts that are longer than a Bachelor's thesis. It is acceptable to include acknowledgements in a Master's thesis, and for books and PhD theses it is standard. Parents and other loved ones are often very flattered to be mentioned in acknowledge-

ments. It is not necessary to mention the supervisor in acknowledgements in texts below the level of a doctoral thesis. The supervisor is mentioned anyway. Supervisors may be embarrassed to be mentioned in the acknowledgements of a Master's thesis (since supervision is simply part of their work), but tastes differ and it is therefore a good idea to discuss this with your supervisor.

Generally speaking, a foreword is not required in a thesis. A foreword usually says something about the reason for the work and about how the work was carried out. It is best to leave this out of a Bachelor's or Master's thesis unless there is something exceptional to report.

3.6 The bibliography

The format for references is described in Section 4.9. Obviously, it is also important to know when references are needed, and to which sources. References are always required when you use other people's ideas or when you use text or illustrations that are not your own. When describing a system, references must be included when you are explaining the various choices made. Choices of this sort must be based on something – preferably a sound knowledge of what other researchers have already tried. Obviously, in your introduction you should include references to similar work, but without interrupting the flow of your account. Sometimes useful, but very tiring to read, are the literature overviews that summarize a great deal of other work but do not really relate it to the author's own work. In the conclusion, after all, full reference can be made to similar conclusions drawn by fellow scientists or academics.

A practical problem that may occur is deciding precisely which source to refer to. Should you refer to the person who originally discovered something, or to the textbook that gives a much clearer explanation of what it is about? Probably the simplest solution is to refer to both. On balance, it is better to over-reference than to under-reference, but be careful about referring to original work that will be difficult for the reader to find. You should be open about the fact that you have not read something, for example by including a reference in the following form: (Schwartz et al., 1997, as cited in Berrah & Laboissière, 1999). In the reference list you then refer to both (Schwartz et al., without the cited bit, and Berrah & Laboissière).

4 Conclusions

Ultimately, value nodes have shown their versatility in performing complex tasks. There are however some massive drawbacks as it is much more computationally heavy to do individual calculations than with matrix operations. In the future with a matrix implementation, it may be better to utilize the Muon optimizer alongside data standardization with synthetic data. With better generalizations with the dataset, we can push this minimalist, bare bones implementation to the limit. In whole, regardless of potential tweaks, this model's ability is not limited by its simplicity.

References

- Berrah, A.-R., & Laboissière, R. (1999). Species: An evolutionary model for the emergence of phonetic structures in an artificial society of speech agents. In J.-D. N. D. Floreano & F. Mondada (Eds.), *Advances in artificial life, lecture notes in artificial intelligence* (Vol. 1674, p. 674-678). Springer Berlin.
- Schwartz, J.-L., Boë, L.-J., Vallée, N., & Abry, C. (1997). Major trends in vowel system inventories. *Journal of Phonetics*, 25, 233-235.